Mahindra University Hyderabad
École Centrale School of Engineering
End-Semester Examination (2024- Batch)
Program: B. Tech. Branch: CM Year: II Semester: Fall
Subject: Real Analysis (MA2104)

Date: 06/12/2025 **Start Time: 10:00 AM**
Time Duration: 3 Hours **Max. Marks: 40**

Instructions:

1. All questions are compulsory.
2. Please start each answer on a separate page.

Q 1: 8 marks

(a) Define a countable set. Let A be the set of all sequences whose elements are 0 and 1. Then show that A is an uncountable set. [4 marks]

(b) Define open and closed sets. Show that a set E is open iff E^c is closed. [4 marks]

Q 2: 10 marks

(a) Define $f : (0,1) \to I\!R$ by $f(x) = \frac{1}{x}$. Show that f is continuous on $(0,1)$ but not uniformly continuous on $(0,1)$. [3 marks]

(b) Give an example of a sequence of continuous functions $\{f_n(x)\}$ on $[0,1]$ such that $\lim_{n\to\infty} f_n(x) = f(x)$ is not a continuous function on $[0,1]$. [3 marks]

(c) Show that a bounded function $f : [a,b] \to I\!R$ is Riemann-Stieltjes integrable iff for each $\epsilon > 0$, there exists a partion $\mathbf{P}$ such that

$$U(f, \mathbf{P}, \alpha) - L(f, \mathbf{P}, \alpha) < \epsilon$$

where $\alpha : [a,b] \to I\!R$ is a monotonically increasing function. [4 marks]

P.T.O.

Q 3: **12 marks**

(a) Define a compact set in $I\!R$. Show that every compact set in $I\!R$ is bounded.

[4 marks]

(b) Let $f : I\!R \to I\!R$ is a continuous function and E is a connected subset of $I\!R$. Then show that $f(E)$ is a connected set in $I\!R$. **[4 marks]**

(c) Let $f : I\!R \to I\!R$ be a continuous function and satisfies

$$f(x+y) = f(x) + f(y) \text{ for all } x, y \in I\!R.$$

Then show that $f(x) = f(1)x$, for all $x \in I\!R$. **[4 marks]**

Q 4: **10 marks**

(a) Give an example of an algebra which is not a σ-algebra. **[3 marks]**

(b) Let A be a subset of $I\!R$. Define a Lebesgue outer measure $\mu^*(A)$. **[3 marks]**

(c) Let E be a measurable set in $I\!R$ and $f, g : E \to I\!R$ are two measurable functions. Then show that $f + g$ is a measurable function. **[4 marks]**

— End of Question Paper —